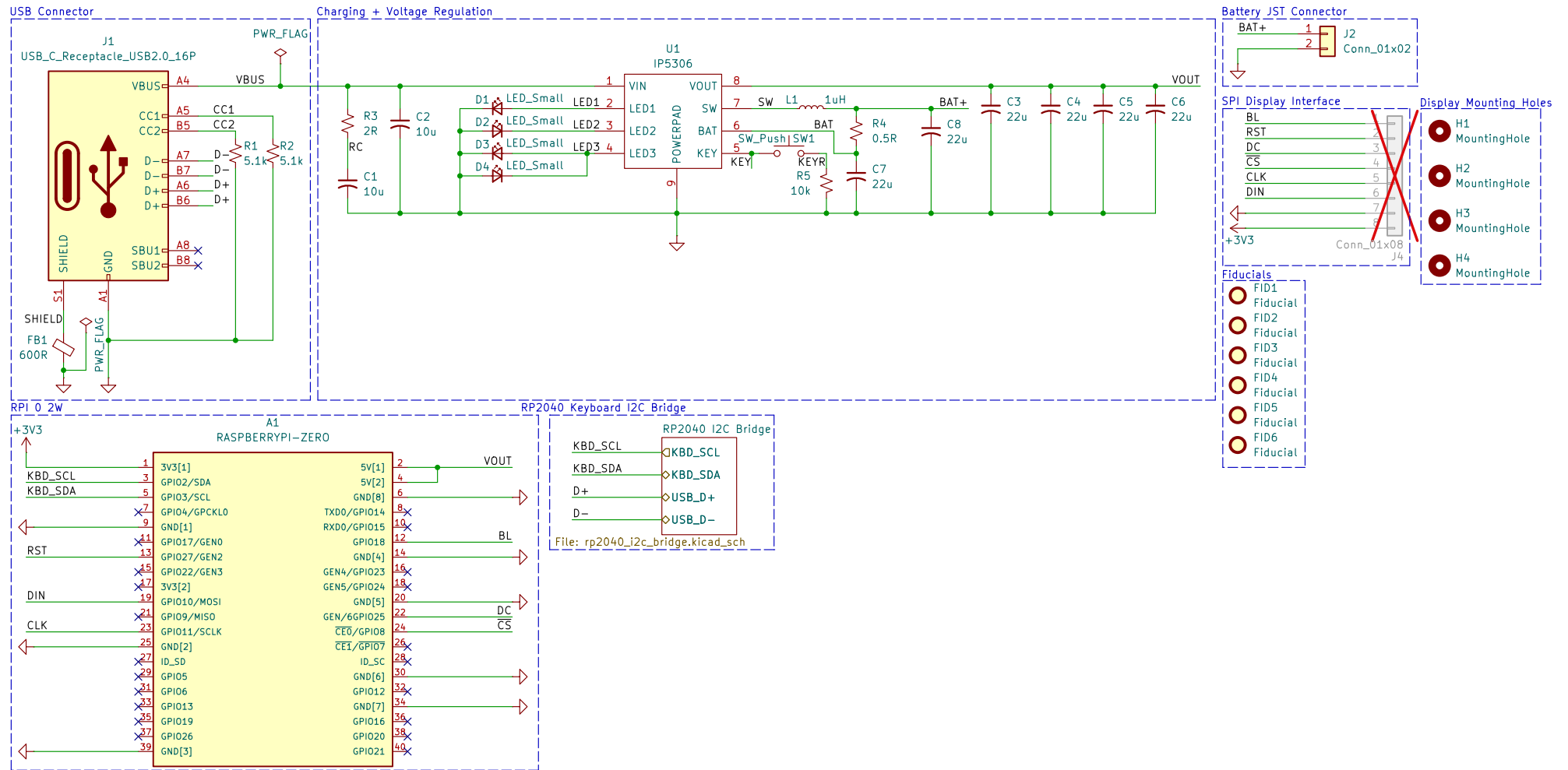




The schematic diagram illustrates the RP2040 hardware design, centered around the RP2040 microcontroller (U2). The design includes a power supply section with a +3V3 input and a +1V1 output, connected to the microcontroller's VDD and VDDIO pins. A series of capacitors (C10-C23) are used for decoupling and filtering. The microcontroller's pins are connected to various peripherals, including USB (USB_DP, USB_DM), QSPI (QSPI_SS, QSPI_SD0-QSPI_SD3, QSPI_SCLK), HSE (HSE_IN, HSE_OUT), and GPIO (GPIO0-GPIO25, GPIO26-ADC0, GPIO27-ADC1, GPIO28-ADC2, GPIO29-ADC3). The diagram also shows connections for KBD_SDA and KBD_SCL, and a TESTEN pin connected to a GND pin.

RP2040 hardware design datasheet's information about impedance matching is based on a 1mm, 2 layer board, but this design is a 1.6mm board. I calculated the differential trace width to be ~0.7mm at 0.2mm spacing with an uninterrupted reference plane on layer 2

U2
RP2040

U4
BBQ10KBD

MIC_VDD COL1 7 COL1
COL2 9 COL2
COL3 10 COL3
COL4 17 COL4
COL5 19 COL5

LEDA_1_2 COL3 10 COL3
LEDA_3_4 COL4 17 COL4
LEDK_1_4 COL5 19 COL5
LEDK_2_3

ROW1 6 ROW1
ROW2 8 ROW2
ROW3 18 ROW3
ROW4 20 ROW4
ROW5 21 ROW5
ROW6 22 ROW6
ROW7 27 ROW7

MIC ROW1 6 ROW1
ROW2 8 ROW2
ROW3 18 ROW3
ROW4 20 ROW4
ROW5 21 ROW5
ROW6 22 ROW6
ROW7 27 ROW7

AGND 4 AGND
GND 1 GND

+3V3 26
+3V3 24
LED 1 4
LED 2 3
LED 3 4
LED 4 5

BL_CONTROL 1
R10 10k
Q1 BSS138

The schematic diagram illustrates the electrical connections for the USB_BOOT pin of the W25Q128JVU. The USB_BOOT pin is connected to a 1k resistor (R8), which is then connected to the CS pin of the W25Q128JVU. A 3V3 supply is connected to the VCC pin of the W25Q128JVU, and a 100nF capacitor (C20) is connected to the VCC pin. The GND pin of the W25Q128JVU is connected to ground. The CS pin is also connected to the 3V3 supply. The diagram includes labels for the components: J3 Conn_01x02, R8 1k, R11 DNF, C20 100n, U3 W25Q128JVU, and the pins: CS, VCC, GND, and the signals: QSPI_SS, QSPI_SCLK, QSPI_SD0, QSPI_SD1, QSPI_SD2, QSPI_SD3, DI/IO0, DO/IO1, WP/IO2, HOLD/RESET/IO3.